

EMERGENCY INNER FORESTAYS

Headsail reefing gear has revolutionised the sail handling on cruising yachts. When weather conditions deteriorate, and you need good pointing ability, the shape of an all purpose furling genoa can be poor. A change down to a small jib requires the furling sail to be unrolled before lowering - increasing the chance of damage to the sail, and danger to the foredeck crew. If the gear has suffered mechanical failure at a critical moment, the yacht may be left bare headed on a lee shore, unable to claw off.

It has yet to be proven that any form of lashing, beads or toggles allow a storm jib to be hoisted over the rolled up genoa in storm conditions. A common approach is to hoist the storm jib "flying" - a dangerous occupation for the foredeck crew as the sail is out of control until fully tensioned. A marginally better solution is to set up a wire halyard as a temporary stay, and hank the storm jib to it. These both rely on an 'elastic' halyard, and standard masthead sheaves. It is difficult to tension them adequately for serious use.

Some propose a fixed or temporary storm stay set somewhere between spreaders and masthead (see Fig.1 below). Any stay set in this position must have some aft support, usually in the form of adjustable or fixed runners. Most cruising yachtsmen avoid runners like the plague!

The emergency stay must be positioned higher than 93% of main forestay height.

The seamanlike, structurally correct yet simple solution is to have an emergency stay connected close to the masthead (see Fig.2). If fitted approximately 300mm below the headbox, it should be clear of the furling headsail gear and very well supported by the backstay. It should also allow any halyard routed through a lead eye to be used with the headsail furling gear or a sail hoisted up the emergency stay.

The emergency stay should be of the same diameter, or one size less than the forestay. For wire sizes up to 8mm, there are special mast attachment plate/hooks which accepts an eye terminal. Above this size, the mast attachment is a normal shroud backing plate or nose tang which accepts a matching wire terminal. These fittings are compatible with most mast types. For the lower stay end, there are many options ranging from simple rigging screws to excellent specialised units incorporating a quick release connection, rigging screw for pre-adjustment, and a Highfield type lever for rapid tensioning. The deck eye connection must be strong - properly linked to a structural bulkhead, or a reinforced deck area tied into the structure below with a wire strop or linkplate.

Fig. 1

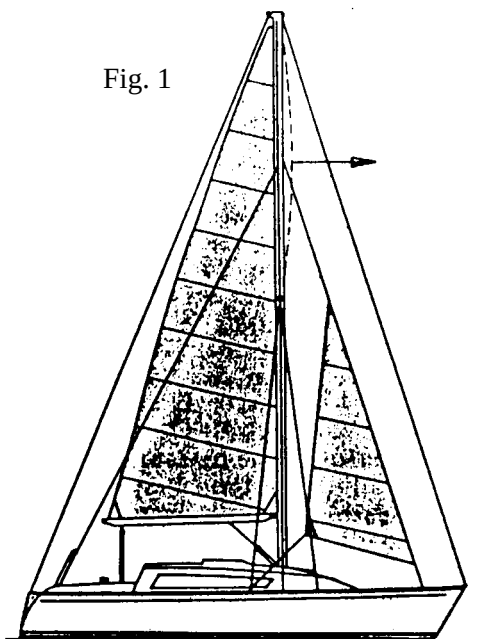
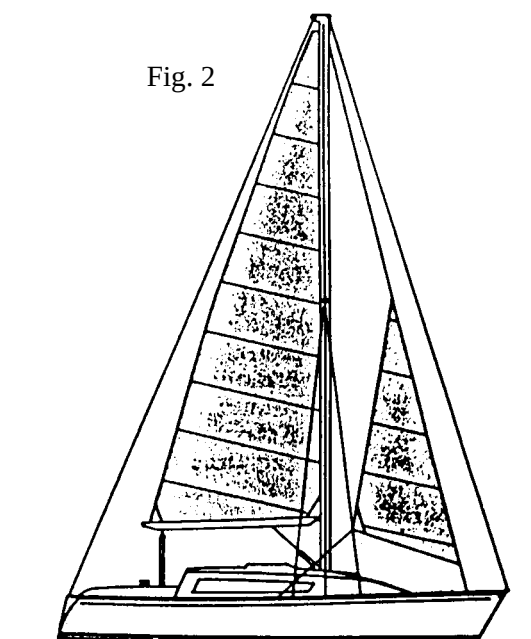


Fig. 2



The illustration opposite shows the main masthead layout features.

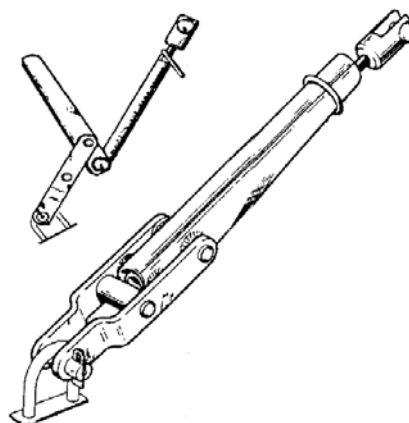
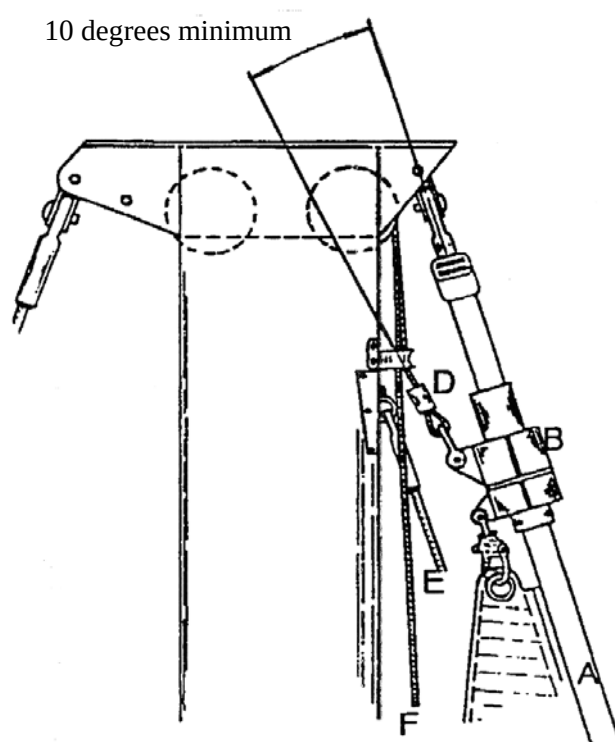
- A Headsail reefing luffspar
- B Sliding halyard swivel
- C Side-by-side bullseye halyard leads (up to 8mm dia., 2 x 508-135-01)
- D Halyard for reefing foresail
- E Inner forestay in operating position
Att. for 5mm dia. (517-904-01)
An. for 6mm dia. (517-905-01)
Att. for 7 & 8mm dia. (517-911-01)
- F Spare foresail halyard for Storm jib

The two bullseye halyard leads are positioned to give a minimum divergence of 10 degrees from the luffspar axis. Either halyard can be used for the reefing sail or any sail fitted to the inner forestay.

Care must be taken over the stay's lower end arrangement. It is essential that the deck attachment in the operating position is strong enough for the task. Of equal importance, in terms of planning, is the stowage method. The stay will spend most of it's life stowed, so it must be held clear of any rig item that could be chafed. The best position is just clear ahead of the spreaders, secured to the deck edge or coachroof side. The stowage should incorporate some means of tensioning the stay when stowed.

The popular options for stowage are:

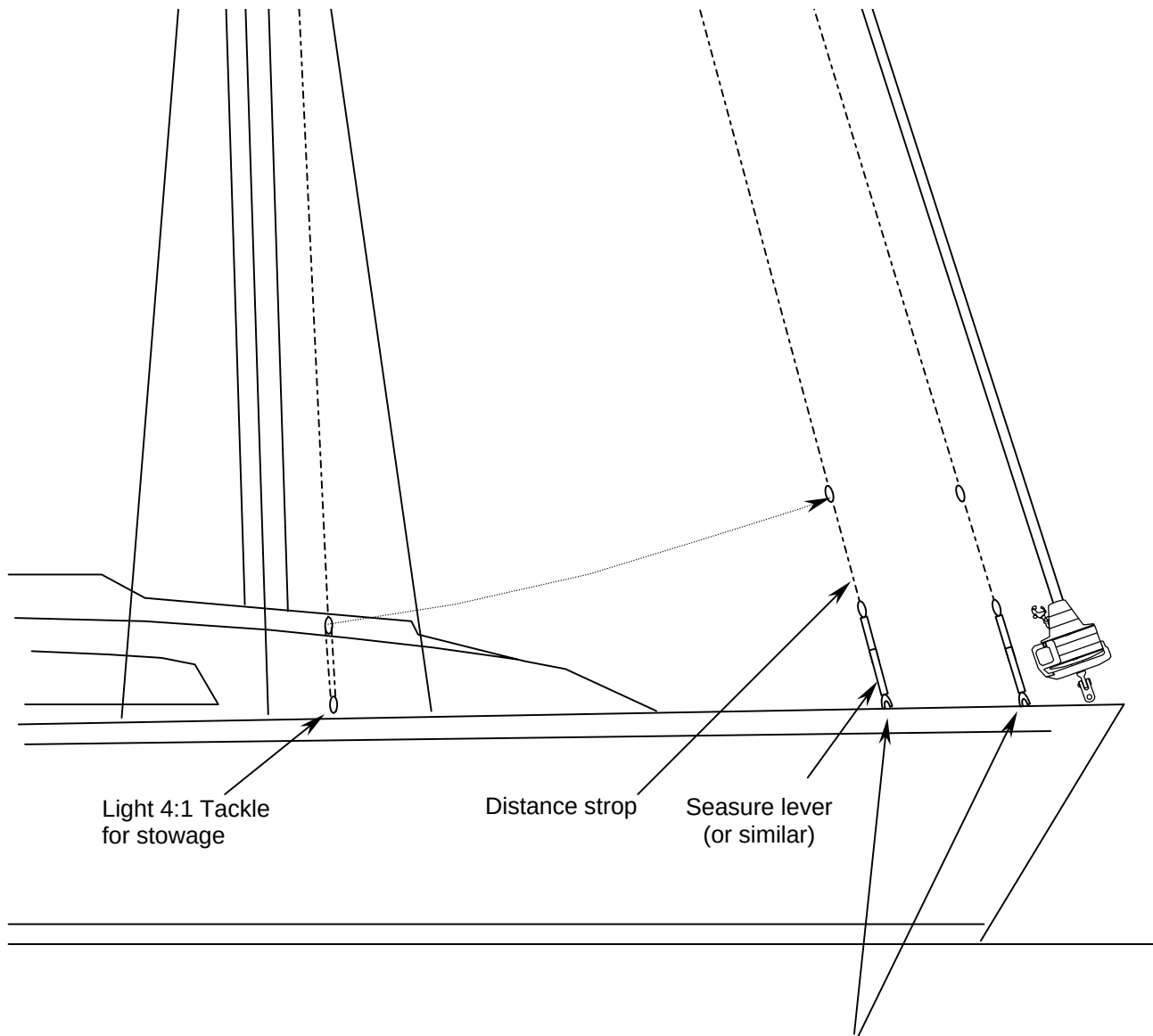
- a. Remove the special adjuster, and tension the stay's lower eye with a small tackle.
- b. Have the stay + adjuster correct length for stowage, and an additional short wire strop or linkplate at the stemhead.
- c. Have a rigging screw or adjuster permanently fitted to the stay. When stowed, the stay will be bent through 90 degrees, and held in place by a lashing secured to the wire above the rigging Screw adjuster



Seasure Quick Release Lower
(for stays up to 8mm diameter)

Emergency Inner Forestay

Deck Layout



Light 4:1 Tackle
for stowage

Distance strop

Seasure lever
(or similar)

Position of lower attachment depends on deck
/ bulkhead structure. The stay can be heavily
loaded, so attachment point must be strong